

Unit and Dimensional Consistency of the PPC Field Force Density Equation

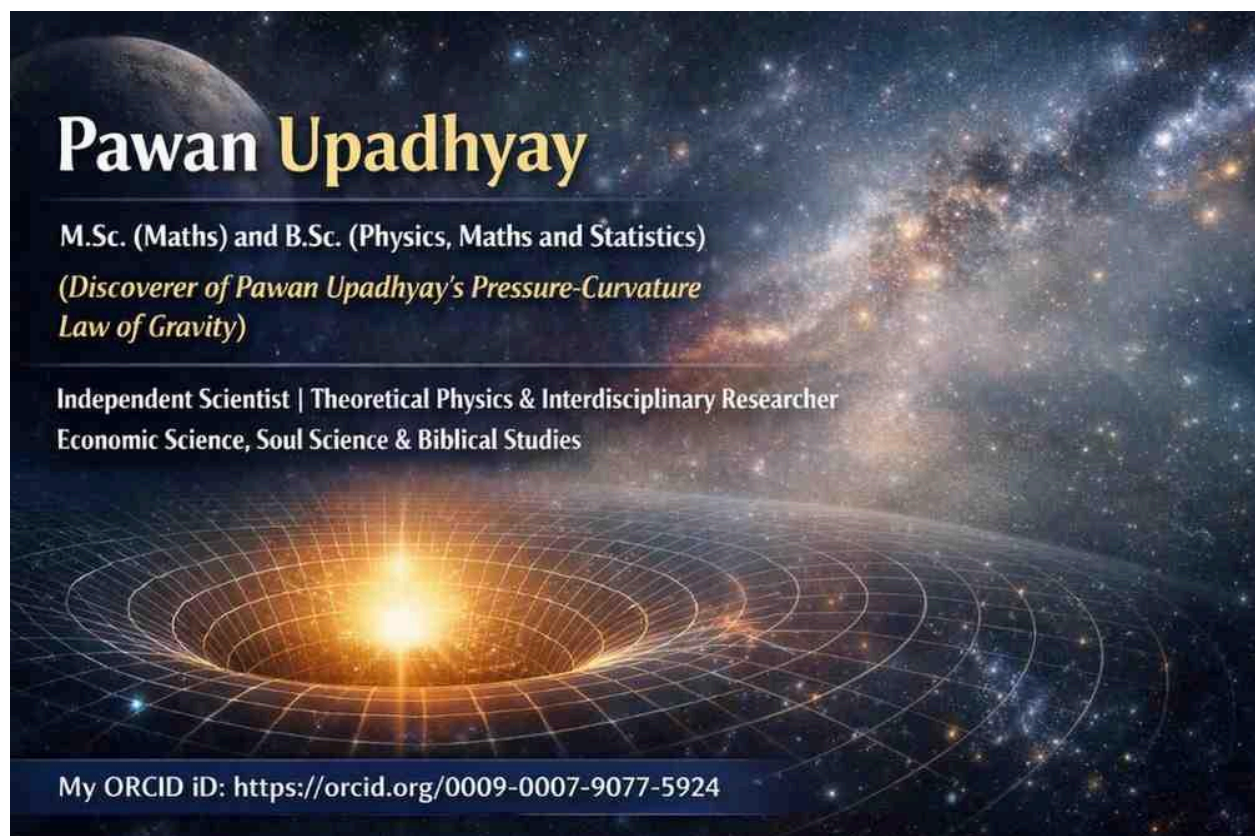
Author: Pawan Upadhyay

Affiliation: Independent Researcher

Email: pawanupadhyay28@hotmail.com

ORCID iD:

<https://orcid.org/0009-0007-9077-5924>



Abstract

Pawan Upadhyay's Pressure Curvature Law of Gravity (PPC Law of Gravity) introduces the concept of a gravitational pressure field P_g , whose spatial variation generates a gravitational field force density. The fundamental field equation is

$$[F = -\nabla P_g]$$

where F denotes field force density and P_g denotes gravitational pressure. This paper examines the dimensional and unit consistency of the equation and demonstrates that the gradient of gravitational pressure naturally possesses the dimensions and SI units of force density. The analysis confirms the mathematical consistency of the PPC field equation.

Abstract

Pawan Upadhyay's Pressure Curvature Law of Gravity (PPC Law of Gravity) introduces the concept of a gravitational pressure field P_g , whose spatial variation generates a gravitational field force density. The fundamental field equation is:

$$F = -\nabla P_g$$

where F denotes field force density and P_g denotes gravitational pressure. This paper examines the dimensional and unit consistency of the equation and demonstrates that the gradient of gravitational pressure naturally possesses the dimensions and SI units of force density. The analysis confirms the mathematical consistency of the PPC field equation.

1. Introduction

Dimensional consistency is a fundamental requirement of every physical theory. For a physical equation to be meaningful, all terms must possess identical dimensions and compatible units.

In the PPC Law of Gravity, gravity is interpreted through the curvature and gradient of a gravitational pressure field. The field force density is defined by

$$[F = -\nabla P_g]$$

where:

- F = gravitational field force density
- P_g = gravitational pressure field
- ∇ = gradient operator

The objective of this paper is to verify the dimensional consistency of this relation.

$$F = -\nabla P_g$$

where:

- F = gravitational field force density
- P_g = gravitational pressure field
- ∇ = gradient operator

The purpose of this paper is to verify the dimensional consistency of this relation.

2. Gravitational Pressure Field

Pressure is defined as force per unit area:

$$[P_g = f/A] \text{ And } P_g = F_p/A$$

where:

- f = force
- A = area
- F_p = surface force
- So $f = F_p$, Both have the same unit Newton.

The SI unit of pressure is

$$[\text{Pa} = \frac{\text{N}}{\text{m}^2}]$$

Since

$$[1 \text{ N} = 1 \text{ kg m s}^{-2}]$$

the dimensions of pressure become

$$[[P_g]$$

$$\frac{M, L, T^{-2}}{L^2}]$$

Therefore,

$$[[P_g]$$

$$M, L^{-1}, T^{-2}]$$

and in SI units

$$[[P_g]$$

$$\text{kg m}^{-1} \text{s}^{-2}]$$

The SI unit of pressure is

$$\text{Pa} = \frac{\text{N}}{\text{m}^2}$$

Since

$$1 \text{ N} = 1 \text{ kg m s}^{-2}$$

the dimensions of pressure become

$$[P_g] = \frac{M L T^{-2}}{L^2} = M L^{-1} T^{-2}$$

Thus,

$$[P_g] = M L^{-1} T^{-2}$$

and the SI unit is

$$\text{kg m}^{-1} \text{ s}^{-2}$$

3. Dimension of the Gradient Operator

The gradient operator is

$$[\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right)]$$

Each spatial derivative contributes one inverse length dimension:

$$[\nabla]$$

$$L^{-1}]$$

3. Dimension of the Gradient Operator

The gradient operator is

$$\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right)$$

Each spatial derivative contributes one inverse length:

$$[\nabla] = L^{-1}$$

4. Dimension of the Pressure Gradient

Applying the gradient operator to the gravitational pressure field gives

$$[\nabla P_g]$$

The dimensions become

$$[[\nabla P_g]]$$

$$[\nabla][P_g]]$$

Substituting dimensions:

$$[[\nabla P_g]$$

$$L^{-1} (M,L^{-1},T^{-2})]$$

Therefore,

$$[[\nabla P_g]$$

$$M,L^{-2},T^{-2}]$$

4. Dimension of the Pressure Gradient

Applying the gradient operator to the gravitational pressure field:

$$\nabla P_g$$

The dimensions become

$$[\nabla P_g] = [\nabla][P_g]$$

Substituting dimensions:

$$[\nabla P_g] = L^{-1}(M L^{-1} T^{-2})$$

Therefore,

$$[\nabla P_g] = M L^{-2} T^{-2}$$

5. SI Unit Analysis

Pressure possesses SI units

$$[P_g = \frac{N}{m^2}]$$

Taking the spatial gradient introduces another inverse meter:

$$[\nabla P_g]$$

$$\frac{N/m^2}{m}]$$

Hence

$$[\nabla P_g]$$

$$\frac{N}{m^3}]$$

Expressed in base SI units:

$$[\frac{N}{m^3}]$$

$$\frac{kg, m, s^{-2}}{m^3}$$

$$kg, m^{-2}, s^{-2}]$$

Thus the pressure gradient has units

$$[kg, m^{-2}, s^{-2}]$$

which is equivalent to

$$[N, m^{-3}]$$

Pressure possesses SI units

$$P_g = \frac{N}{m^2}$$

Taking the spatial gradient introduces another inverse meter:

$$\nabla P_g = \frac{N/m^2}{m} = \frac{N}{m^3}$$

Expressed in base SI units:

$$\frac{N}{m^3} = \frac{kg\,m\,s^{-2}}{m^3} = kg\,m^{-2}\,s^{-2}$$

Thus, the pressure gradient possesses units

$$N\,m^{-3}$$

or equivalently

$$kg\,m^{-2}\,s^{-2}$$

6. Definition of Field Force Density

Force density is defined as force per unit volume:

$$[F = f/V]$$

where

$$[V = m^3]$$

Therefore,

$$[F]$$

$$\frac{N}{m^3}]$$

Using base SI units,

$$[F]$$

$$kg, m^{-2}, s^{-2}]$$

and in dimensional form

$$[F]$$

$$M, L^{-2}, T^{-2}]$$

Force density is defined as force per unit volume:

$$F = \frac{f}{V}$$

where

$$V = m^3$$

Therefore,

$$[F] = \frac{N}{m^3}$$

Using base SI units,

$$[F] = kg\,m^{-2}\,s^{-2}$$

and in dimensional form

$$[F] = M\,L^{-2}\,T^{-2}$$

7. Consistency Verification

The left-hand side of the PPC equation has dimensions

$$[[F]$$

$$M, L^{-2}, T^{-2}]$$

The right-hand side has dimensions

$$[[\nabla P_g]$$

$$M, L^{-2}, T^{-2}]$$

Thus,

$$[[F]$$

$$[\nabla P_g]]$$

Likewise, the SI units satisfy

$$[\frac{N}{m^3}$$

$$\frac{N}{m^3}]$$

Therefore,

$$[F = -\nabla P_g]$$

is dimensionally and unit consistent.

The left-hand side of the PPC equation has dimensions

$$[F] = M L^{-2} T^{-2}$$

The right-hand side has dimensions

$$[\nabla P_g] = M L^{-2} T^{-2}$$

Thus,

$$[F] = [\nabla P_g]$$

Likewise, the SI units satisfy

$$\frac{N}{m^3} = \frac{N}{m^3}$$

Therefore,

$$F = -\nabla P_g$$

is dimensionally and unit consistent.

8. Physical Interpretation

The PPC equation states that a spatial variation in gravitational pressure generates a force density within the gravitational field.

The negative sign indicates that the field force density points toward decreasing gravitational pressure:

$$[F \propto -\nabla P_g]$$

This structure is mathematically analogous to pressure-gradient forces in continuum mechanics and fluid dynamics.

8. Physical Interpretation

The PPC equation states that a spatial variation in gravitational pressure generates a force density within the gravitational field.

The negative sign indicates that the field force density points toward decreasing gravitational pressure:

$$F \propto -\nabla P_g$$

This mathematical structure is analogous to pressure-gradient forces encountered in continuum mechanics and fluid dynamics.

9. Conclusion

The dimensional analysis confirms that the PPC field equation

$$[F = -\nabla P_g]$$

is mathematically self-consistent. The principal results are:

1. Gravitational pressure has dimensions M, L^{-1}, T^{-2} .
2. The gradient operator contributes L^{-1} .
3. The pressure gradient has dimensions M, L^{-2}, T^{-2} .
4. These dimensions are identical to those of force density.

5. Both sides possess SI units N, m^{-3} .

The dimensional analysis confirms that the PPC field equation

$$F = -\nabla P_g$$

is mathematically self-consistent.

The principal results are:

1. Gravitational pressure has dimensions M, L^{-1}, T^{-2} .
2. The gradient operator contributes L^{-1} .
3. The pressure gradient has dimensions M, L^{-2}, T^{-2} .
4. These dimensions are identical to those of force density.
5. Both sides possess SI units N, m^{-3} .

Therefore, the PPC field-force-density equation satisfies the requirements of dimensional analysis and unit consistency, supporting its formulation as a continuum gravitational field relation.

Notes

To avoid ambiguity in PPC literature, it is useful to distinguish:

[f = ordinary force]

with SI unit

[[N]]

and

[F = field force density]

with SI unit

[[N,m⁻³]]

For clarity in PPC literature, it is useful to distinguish between ordinary force and field force density:

f = ordinary force

with SI unit

$[N]$

and

F = field force density

with SI unit

$[N\ m^{-3}]$

This notation clearly separates force from force density in PPC field equations and weak-field formulations.